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SUPERSEDING
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MILITARY SPECIFICATION

PLATE SET, PSEUDOISOCHROMATIC, 15 PLATE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of vision test set, suitable for use in determining red-green color-vision deficiencies.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6515

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3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout, suitable for the purpose intended, and as hereinafter specified.

3.2 Style and design. The color-vision test set shall consist essentially of one demonstration and 14 diagnostic pseudoisochromatic plates, a looseleaf ring binder, and instructions for use. Test set shall be as developed by the Armed Forces-National Research Council Committee on Vision, or approved equal.

3.2.1 Pseudoisochromatic plates. Plates shall be produced on dull-white, lightweight cardboard to show Arabic numerals, arranged in circular dots of varying size upon a circular or rectangular background of dots similar in size and distribution, but having tints of colors that are likely to be confused by individuals having defective color perception. Plates having rectangular backgrounds of dots shall measure 4-1/2 inches wide (plus 1/16 inch, minus 0 inch) by 3-3/4 inches high (plus 1/8 inch, minus 0 inch), the test portions measuring 4-3/16 inches wide (plus 1/16 inch, minus 0 inch) by 3-1/2 inches high (plus 1/16 inch, minus 0 inch). Plates having circular backgrounds of dots shall be approximately square, shall measure 4 inches (plus or minus 5/16 inch) on the sides, and the test portions shall be circular, 3-3/8 inches to 3-9/16 inches in diameter. The numerals, designs, and coloring of the plates shall be such as to demonstrate quickly and conclusively defective red-green vision. Each plate shall be securely and neatly mounted or printed upon black heavy-weight paper or lightweight cardboard pages. Each page shall measure 5-3/4 inches wide (plus or minus 1/4 inch) by 6-5/8 inches high (plus or minus 1/4 inch). The left side of each page (i.e., the long edge to the left with the front surface toward the viewer and the numerals in the test plate upright) shall be perforated to fit the looseleaf binder hereinafter specified. The centers of the perforations and the size and spacing of the perforations shall be such that the pages may be turned freely without binding on the rings.

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3.2.1.1 Plate numbers. The colors and size distribution of the dots forming the backgrounds; the form of the background (rectangular or circular); the colors, numbers, and distribution of the dots forming the numerals in the demonstration plate; and the 14 diagnostic plates comprising the set shall be as in either of the following columns:

	A 1/	B 2/
Demonstration plate	25	1
Diagnostic plates	6	2
	8	3
	9	4
	12	5
	14	6
	19	7
	20	8
	23	9
	24	10
	27	11
	29	12
	41	13
	42	14
	43	15

1/ Plate numbers (not to be confused with normal responses for these plates, i.e., the numbers appearing in the test portions of the plates. For example, the number 12 appears in the test portion of plate number 25 - the Demonstration Plate) as shown in the 1940 Edition of the American Optical Co.'s "Pseudoisochromatic Plates for Testing Color Perception."

2/ Plate Sequence Numbers as appearing on "Sample Record Forms" included in "Dvorine Pseudo-Isochromatic Plates," and Second Edition, Baltimore, Maryland, 1953.

3.2.2 Binder. The binder for holding the demonstration and diagnostic plates shall be of the multi-ring looseleaf type with hard covers suitably and durably bound. The binder shall be of such size and design as to extend beyond the enclosed pages of the set and protect these pages against damage, when the booklet is closed. The ring assembly mounted to the back of the book shall have a locking (and opening) mechanism operable by means of appropriate handles at both ends of the bar with not less than six rings for holding the pages (plates). The rings shall be rectangular or round in section and of sufficient inside diameter to allow the pages to be turned easily and freely. The outside surface of the book shall be blue or gray.

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3.2.3 Instructions for use, Instructions for use shall be printed on heavy-stock material similar to the material used for the pages for the test plates. The page(s) containing the instructions shall be incorporated in front or at the end of the test book and shall cover, as a minimum, the following points:

A. Physical arrangements.

1. Description of the test set.

2. Care when not in use.

3. Light source and lighting arrangements to be used.

4. Subject location relative to plates.

B. Administration.

1. Examiner's instructions to subject.

2. Procedure in examination, including initial showing of test plate, time each diagnostic plate is shown, examiner's actions in event of hesitation etc., by subject.

3. Provisions to circumvent learning of plates by subject.

C. Scoring.

1. Statement of level (number of plates read correctly) at which subject is rated "Color Vision Normal", or numbers of plates read incorrectly at which subject is rated "Color Vision Defective". (Both statements may be included.)

2. Scoring (correct or incorrect) for plates with two-digit numbers, or in instances of no response from subject.

3. Relation of light source, subject distance, and timing to interpretation of results.

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D. Purpose. Statement of purpose of test.

NOTE: A draft of proposed instructions prepared in accordance with the foregoing shall be submitted by the bidder to, and approved by, the procuring agency prior to award of contract or purchase order, unless such approval had been received after 1 January 1959. In those cases where such approval had been received, the bidder shall forward to the procuring activity a certificate to the effect that the instructions for use that are being furnished in the booklets are identical to the instructions previously approved. In addition, when inspection is performed at source, a copy of the certificate shall be furnished to the cognizant Government inspector.

3.2.4 Slipcase. The test set shall be furnished in a suitable cardboard slipcase. The long edges of the case forming the opening shall have finger cutouts to permit grasping the binder for easy removal. The dimensions of the slipcase shall not exceed 7-1/4 inches wide by 7-3/4 inches high by 1-9/16 inches deep.

3.3 Identification marking. The outside surface of the top of the binder shall be permanently and legibly marked with the name or registered trademark of the contractor, the letters U.S., and the Federal stock number. Commercial identification of the item may be included.

3.4 Workmanship. Workmanship shall be first class throughout. Color-vision test sets shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditiory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.3 Examination. The plate sets shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) plate set, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

Table I. Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	48 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-636. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix of PPP-B-636. In addition, waterproofing shall be as specified in 30.4.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

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5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-636. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix of PPP-B-636. In addition, waterproofing shall be as specified in 30.4.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

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